

# Predicting the Working Life of Algorithmically Discovered Trading Strategies

*A survival-analysis meta-model, built and verified against synthetic ground truth*

|                   |                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------|
| Author            | Robert Stevenson                                                                           |
| Repository        | strategy-survival-model                                                                    |
| Run               | seed 7, 5,000 strategies, 5 temporal folds                                                 |
| Source of figures | reports/metrics.json, regenerated by <code>python scripts/run_synthetic_pipeline.py</code> |

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## 1. Summary

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An automated strategy search produces a queue of candidates that all clear validation. They stop working at very different rates, and the rate determines how much capital a new strategy should get and when it should be reviewed. This report covers a model that predicts how long a newly discovered strategy will keep working, using only the metadata recorded on the day it is deployed.

The finding that matters more than the model's accuracy is that the primary metric allocators implicitly rank on, validation Sharpe, is worse than useless for this question. Ranking strategies by validation Sharpe scores 0.410 on a concordance index where 0.500 is a coin flip, and it stays below 0.500 in all 5 folds. The cause is selection. Every strategy entered the queue by clearing a Sharpe threshold, so beyond that threshold a high score reflects overfitting more often than edge, and overfit strategies decay fastest. In this report the inversion is demonstrated on synthetic data whose generator installs that selection step by construction; section 3 walks the mechanism, and whether it holds on any particular real book is an empirical question this report does not settle.

A model reading walk-forward consistency, how a strategy's returns held up across repeated refit-and-test steps through its own validation history, instead reaches 0.781 (95% bootstrap interval 0.772 to 0.790, pooled over 3,000 out-of-fold strategies). Because the data is synthetic but includes built-in baseline noise, the best score any model could achieve is computable via an oracle model, and it is 0.820. The winning valid model sits 0.038 below that ceiling, so most of the remaining error is irreducible noise rather than model capacity.

The same pipeline also runs on real data. A companion report in this repository applies it to 262,763 public City of Chicago business licences and reaches a pooled out-of-fold concordance of 0.697

( `reports/chicago_demo/` ). Most of that figure reflects licence-category membership rather than ranking within a category, where comparisons score 0.544; the companion report gives the decomposition.

Note: the methodology in this report is built and verified against known ground truth derived from the synthetic data. No production metadata is presented here. Its purpose is to demonstrate methodology, not strategy edge.

## 2. The problem

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Strategies discovered by an automated search decay. An edge that clears validation today is usually unprofitable within months, and lifetimes vary by an order of magnitude among strategies with identical headline metrics.

Three operational decisions depend on the decay rate: how much capital a new strategy receives on day one, when its first serious review is scheduled, and when it is retired. The default is to treat every new strategy the same, which misallocates in both directions. It overfunds strategies that will not survive the quarter and underfunds the ones that would have run for a year.

The question is whether discovery-time metadata carries enough signal to separate them. Strategies die for causes that leave traces in that metadata. A strategy selected from a hundred thousand candidates carries more selection bias than one selected from a hundred. A strategy whose walk-forward Sharpe was already sliding during validation is saying something. None of that is visible in a single validation statistic, and all of it is already recorded.

## 3. Why ranking by validation Sharpe fails

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Every strategy in the population cleared a validation-Sharpe threshold of 0.8, because that is how a strategy enters a deployment queue. An observed Sharpe is the sum of true edge, an overfitting component, and measurement noise. Conditioning on that sum exceeding a threshold means the survivors with the highest observed scores are disproportionately the ones whose overfitting and noise happened to break upward.

Past the threshold, additional observed Sharpe is more likely inflation than edge. Because the inflated strategies are the overfit ones, and overfit strategies decay fastest, higher validation Sharpe actively predicts *shorter* working life. The result is not a weak predictor but an inverted one, at 0.410 pooled, ranging from 0.354 to 0.427 across folds and never once above the 0.500 of a coin flip. An inverted ranking is still

information, since it can be read backwards. Ranking by low validation Sharpe scores 0.590 pooled, well short of the 0.781 the model reaches, so the reversal recovers some signal but does not substitute for the model.

This is what makes the modeling problem worth posing. If validation Sharpe alone could rank survival correctly there would be nothing to add. A meta-model earns its place precisely because the selection process has already consumed the obvious signal.

## 4. Data

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The run reported here draws 5,000 strategies with seed 7, with initial discovery dates between 2021-07-01 and 2026-06-01, observed to 2026-07-01. Median observed lifetime is 91 days. 6.8% of strategies are right-censored, meaning the end of their working life was never observed: they are either still running at the observation cutoff or administratively retired, a retirement the generator draws for 6% of strategies independent of performance. Censoring concentrates among recently discovered strategies, which have had the least time to fail before the cutoff, so the final fold's test block is far more censored than the population overall. Survival time is log-normal in the latent quantities with a scale of 0.55 on log-days, which is the noise that places the ceiling below a perfect score.

The data is synthetic. Addendum A describes the generating process, what it deliberately builds into the population, and the two verification checks that a known ground truth makes possible.

## 5. Method

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### 5.1 Model class

The target is a duration with incomplete observations, so the model is an accelerated failure time (AFT) model, implemented as XGBoost with the `survival:aft` objective. Censoring enters through interval labels, where an observed death is the interval  $[t, t]$  and a censored strategy is  $[t, \text{infinity})$ .

Regression on lifetime was rejected because censored rows have no target, and dropping them discards the longest-lived strategies, which biases the model toward pessimism. Classification at a fixed horizon handles censoring awkwardly and answers only one question. AFT keeps every row and returns a full time distribution, which collapses to any horizon an allocator asks about.

Concretely, the fitted model predicts one number per row, the median survival time in days. The full curve comes from wrapping a log-normal distribution of fitted width around that median, so the probability of

surviving any horizon is a point read off that curve. The width is a single value shared by every row; how it is selected and calibrated is covered in section 5.3.

The synthetic feature matrix is finite by construction, so the model does no imputation on this data.

## 5.2 Temporal validation and label re-censoring

Evaluation uses 5 expanding-window folds ordered by discovery date. The earliest 40% of strategies is burn-in that is only ever trained on. Each fold trains on every strategy discovered before its split date and tests on the next 600 strategies, so training sets grow from 1,999 to 4,399 rows.

Training labels are re-censored at each split date. A strategy discovered two years before a split may have died three months after it, and its recorded label contains that future death. Any model trained at that split date could only have known the strategy was still running, so every post-split death is rewritten as a censoring at the split. Omitting this step raises measured scores while silently importing future information, which makes it the most consequential detail in the pipeline. It is implemented as a standalone function ( `cv.recensor` ) with dedicated tests, because it transfers unchanged to any duration problem on operational data.

Test labels use the full observation window, because the restriction exists to keep the future away from the model, not away from the scorer.

## 5.3 Hyperparameter selection and calibration

Hyperparameters are selected once, on the first fold's training window, using an inner temporal split and held-out censored log-likelihood. Likelihood rather than concordance drives selection, and the reason is a failure this pipeline recorded rather than avoided.

An earlier version selected on concordance and looked correct: ranking metrics landed where they ultimately did. It then lost to a no-skill reference forecast on 365-day Brier score, the mean squared error of a probability forecast. A concordance index is invariant to the predictive scale, so the search had no reason to prefer a usable distribution width and settled on one far too wide. Nothing in the training loop objected, because nothing in the training loop was measuring it. Selection now uses likelihood, which penalizes a broken scale, and the predictive log-normal scale is fitted separately on a temporal tail slice the early-stopping probe never trained on. The selected loss scale was 0.6, and the calibrated predictive scale came out at 0.64.

The general lesson holds beyond this project. A model can rank well and misstate probabilities at the same time, and the only defense is evaluating both.

## 5.4 Baselines

Three references bound the result. A Cox proportional hazards model on the same features is the standard survival alternative and answers whether the gradient-boosted model was necessary. Ranking by validation Sharpe is the heuristic a backtest-driven allocator applies implicitly. The oracle ranks by the latent quantity that generated the lifetimes and gives the ceiling that measurement noise imposes.

# 6. Results

## 6.1 Discrimination

Pooled over 3,000 out-of-fold strategies, with 95% percentile bootstrap intervals over 500 resamples:

**Table 1.** Concordance index by method, pooled across folds. Higher is better; 0.500 is a coin flip.

| METHOD                               | CONCORDANCE (HARRELL'S C) | 95% INTERVAL          |
|--------------------------------------|---------------------------|-----------------------|
| Oracle on latent log-time (ceiling)  | 0.820                     | not resampled         |
| <b>XGBoost AFT (pooled)</b>          | <b>0.781</b>              | <b>0.772 to 0.790</b> |
| XGBoost AFT (fold mean)              | 0.781                     | not resampled         |
| Cox proportional hazards (fold mean) | 0.781                     | not resampled         |
| Rank by validation Sharpe            | 0.410                     | 0.397 to 0.422        |

Two results in that table need stating plainly rather than being left to inference.

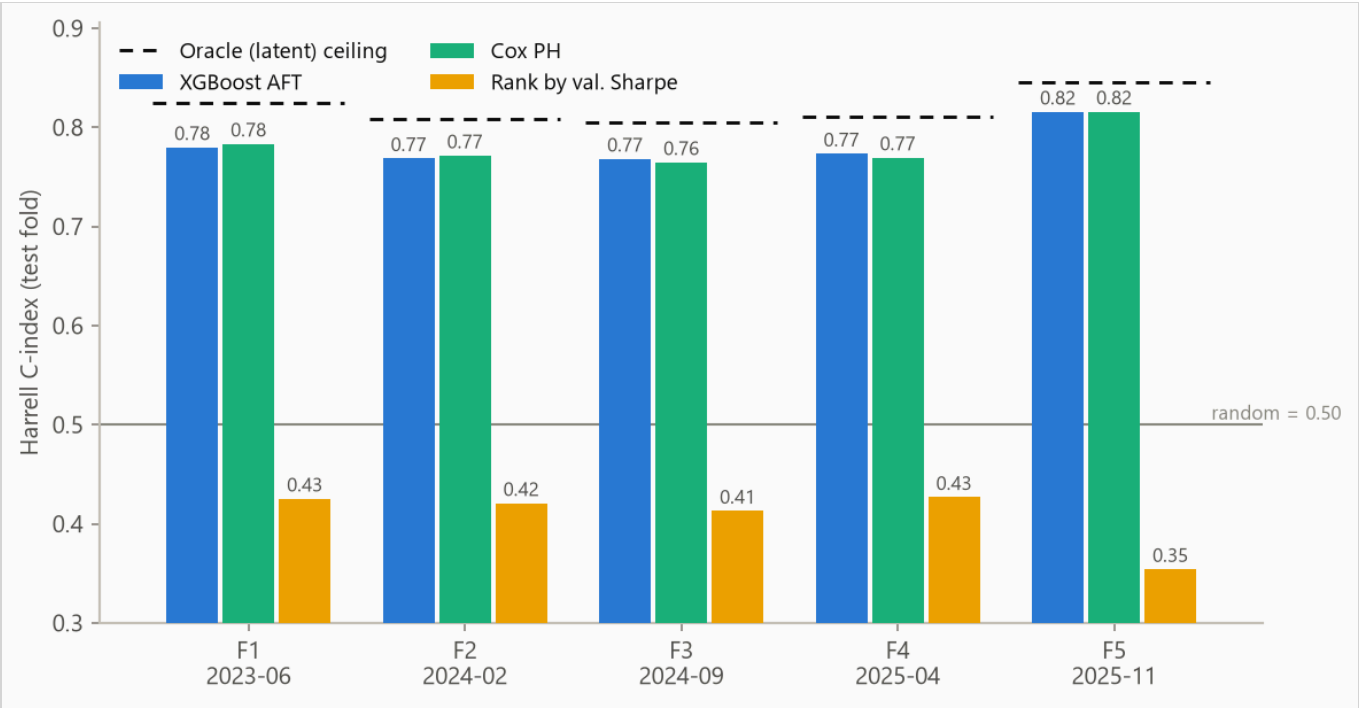
The boosted model ties the Cox baseline, 0.781 against 0.781. Both figures are fold means, which is the only like-for-like basis: each fold refits Cox, and its risk scores are relative ones centred on that fold's own training window, so they carry no common scale across folds and pooling them would compare different units. The generator's observable structure is close to additive, and 1,999 to 4,399 training rows is not enough for trees to find much beyond what a penalized linear model in the log-hazard already captures. I report the tie rather than adding interactions to the generator until the headline model wins, because a benchmark tuned until it loses is not a benchmark. The tie is itself informative. On this problem a simpler model suffices, and knowing that is worth more than an engineered margin.

Both models sit about 0.04 below the oracle. The gap between 0.781 and a perfect score is therefore mostly unpredictable variation in when a strategy actually stops working, not unused signal.

**Table 2.** Per-fold results. Censoring is the share of each fold's test strategies still running at the observation cutoff.

| FOLD | SPLIT DATE | TRAIN N | TEST N | CENSORED | AFT   | COX   | SHARPE | ORACLE |
|------|------------|---------|--------|----------|-------|-------|--------|--------|
| 1    | 2023-06-22 | 1,999   | 600    | 2%       | 0.780 | 0.783 | 0.425  | 0.824  |
| 2    | 2024-02-03 | 2,600   | 600    | 2%       | 0.769 | 0.771 | 0.421  | 0.808  |
| 3    | 2024-09-01 | 3,199   | 600    | 2%       | 0.768 | 0.765 | 0.413  | 0.805  |
| 4    | 2025-04-03 | 3,800   | 600    | 9%       | 0.773 | 0.769 | 0.427  | 0.811  |
| 5    | 2025-11-03 | 4,399   | 600    | 39%      | 0.816 | 0.816 | 0.354  | 0.846  |

Figure 1 plots the per-fold concordance from Table 2; each fold's censoring mix is in the table.



**Figure 1.** Concordance by fold. AFT performance is stable from 0.768 to 0.816. The final fold scores highest, but 39% of its test strategies are censored against 2% in the first fold, which changes the set of comparable pairs. Concordance is not strictly comparable across folds with different censoring mixes, so this plot supports a claim of stability rather than improvement.

## 6.2 Calibration

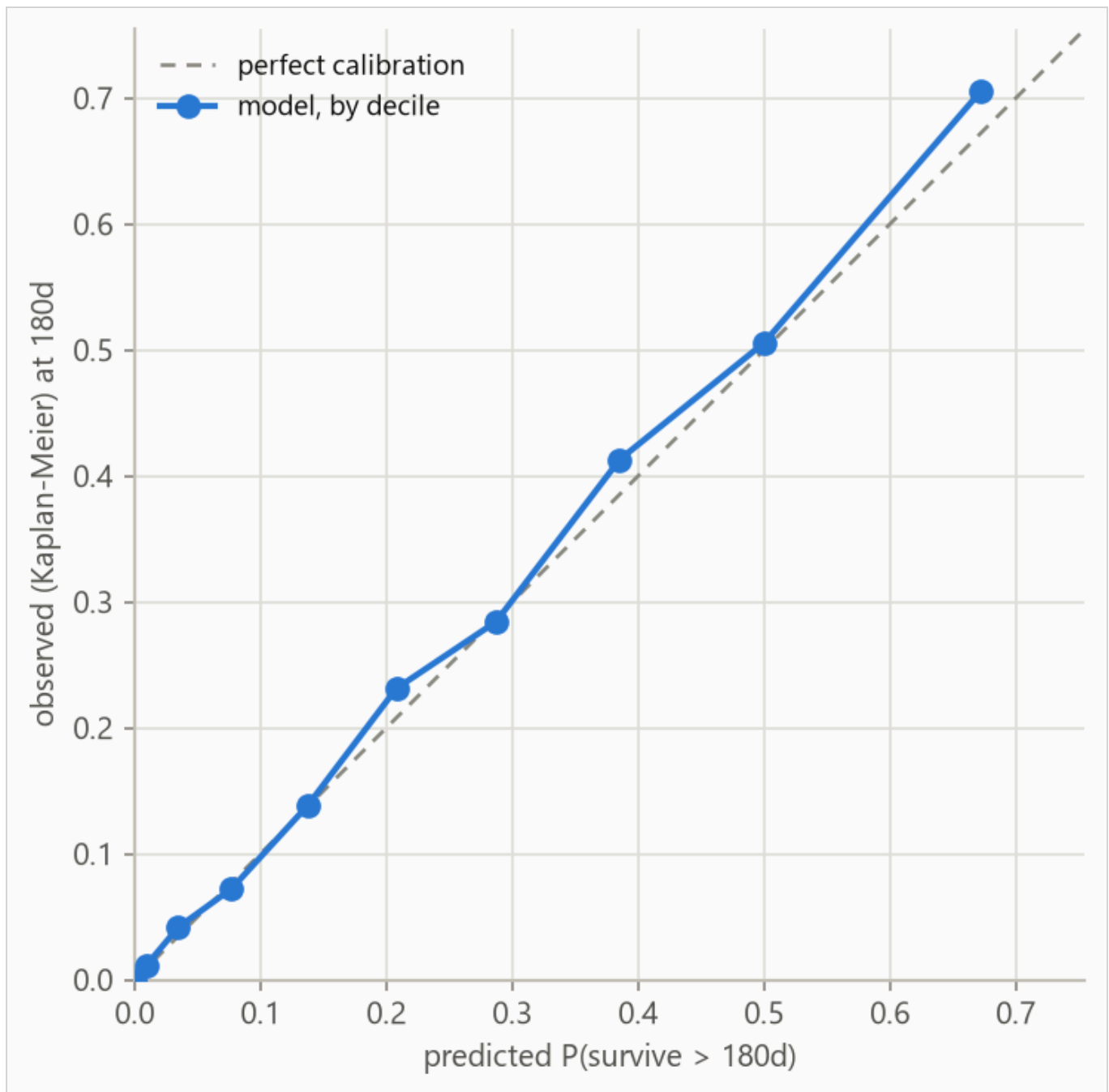
Predicted median survival times are converted to survival probabilities under the calibrated log-normal, then checked two ways. The Brier score is the mean squared error of a probability forecast, lower being better, and here it is weighted by the inverse probability of censoring (IPCW), so that strategies whose outcome was censored away do not bias the result. The no-skill reference assigns every strategy the pooled Kaplan-Meier

marginal, the population's own survival curve with censored strategies counted for as long as they were observed, ignoring all features.

**Table 3.** IPCW Brier score by horizon. Lower is better.

| HORIZON  | AFT   | COX   | NO-SKILL MARGINAL |
|----------|-------|-------|-------------------|
| 90 days  | 0.148 | 0.150 | 0.249             |
| 180 days | 0.132 | 0.132 | 0.182             |
| 365 days | 0.050 | 0.051 | 0.056             |

The margin over the no-skill reference is wide at 90 days and narrow at 365. That is the expected shape. Median survival is about 91 days, so by one year nearly every strategy has stopped working and the marginal forecast is already close to certain. Little skill remains to demonstrate at that horizon.



**Figure 2.** Predicted against observed survival at 180 days, by predicted decile. Observed frequencies are Kaplan-Meier estimates within each bin, so censored strategies contribute correctly rather than being dropped. The largest deviation is 0.033 in decile 10.



**Table 4.** Decile calibration at 180 days, the values plotted in Figure 2.

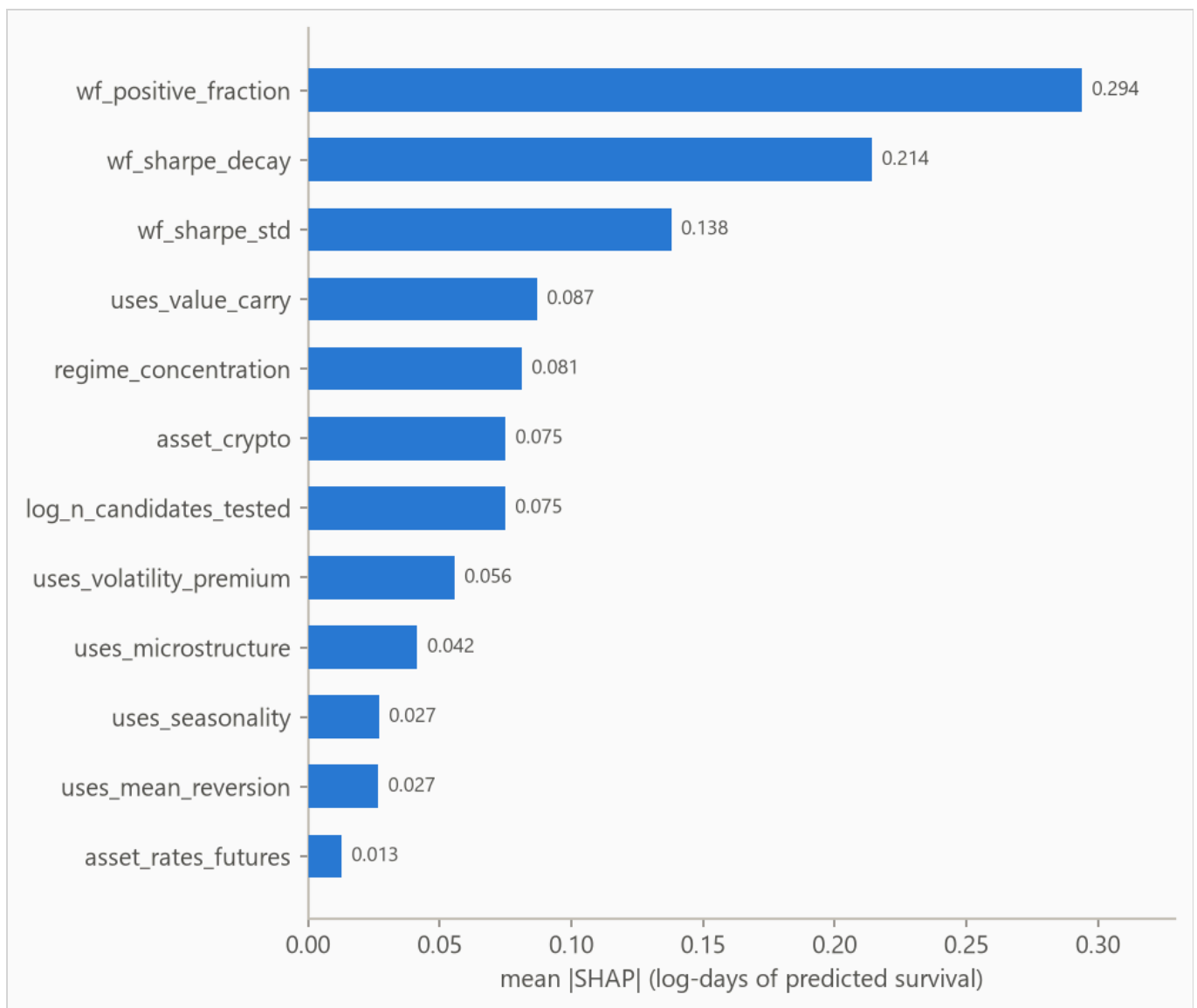
| DECILE | N   | PREDICTED | OBSERVED (KM) | DEVIATION |
|--------|-----|-----------|---------------|-----------|
| 1      | 300 | 0.001     | 0.000         | -0.001    |
| 2      | 300 | 0.010     | 0.012         | +0.002    |
| 3      | 300 | 0.035     | 0.042         | +0.007    |
| 4      | 300 | 0.077     | 0.072         | -0.005    |
| 5      | 300 | 0.138     | 0.139         | +0.001    |
| 6      | 300 | 0.208     | 0.231         | +0.023    |
| 7      | 300 | 0.287     | 0.284         | -0.003    |
| 8      | 300 | 0.385     | 0.413         | +0.028    |
| 9      | 300 | 0.500     | 0.506         | +0.006    |
| 10     | 300 | 0.672     | 0.705         | +0.033    |

One caveat on the decile table. The observed value inside each decile is a Kaplan-Meier estimate, and when a decile's last observed strategy falls short of the 180-day horizon the estimate carries its last value forward. In small or heavily censored deciles the observed column is softer than its three decimals look, and deviations there should be read accordingly.

## 7. What the model uses

Feature attributions (SHAP values) are computed on the log-time margin, so a value of +0.3 multiplies predicted survival time by about 1.35 and negative values shorten it.

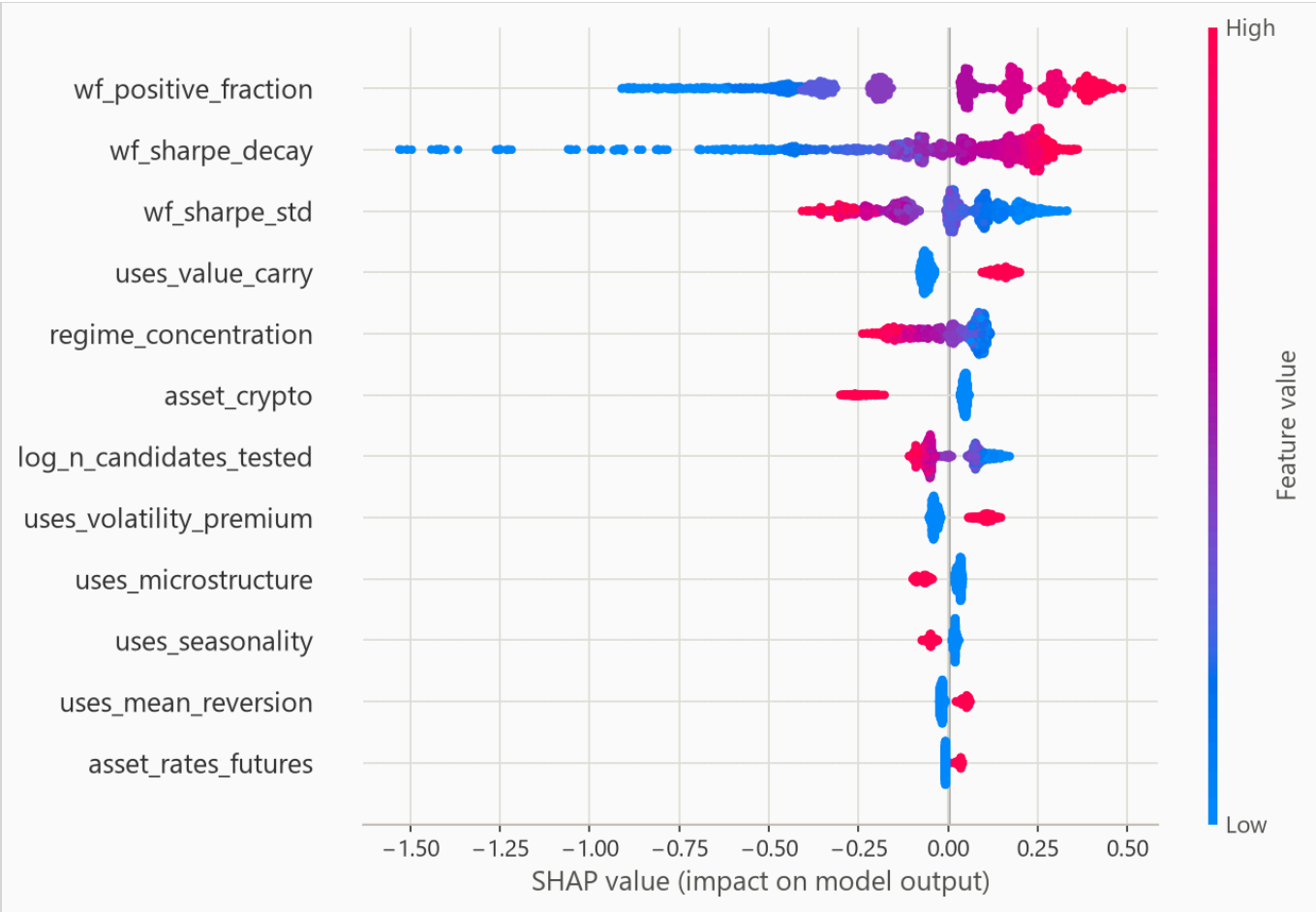
Figure 3 ranks features by mean absolute attribution, Table 5 lists the top eight, Figure 4 shows the per-row spread, and Figure 5 traces attribution against feature value.



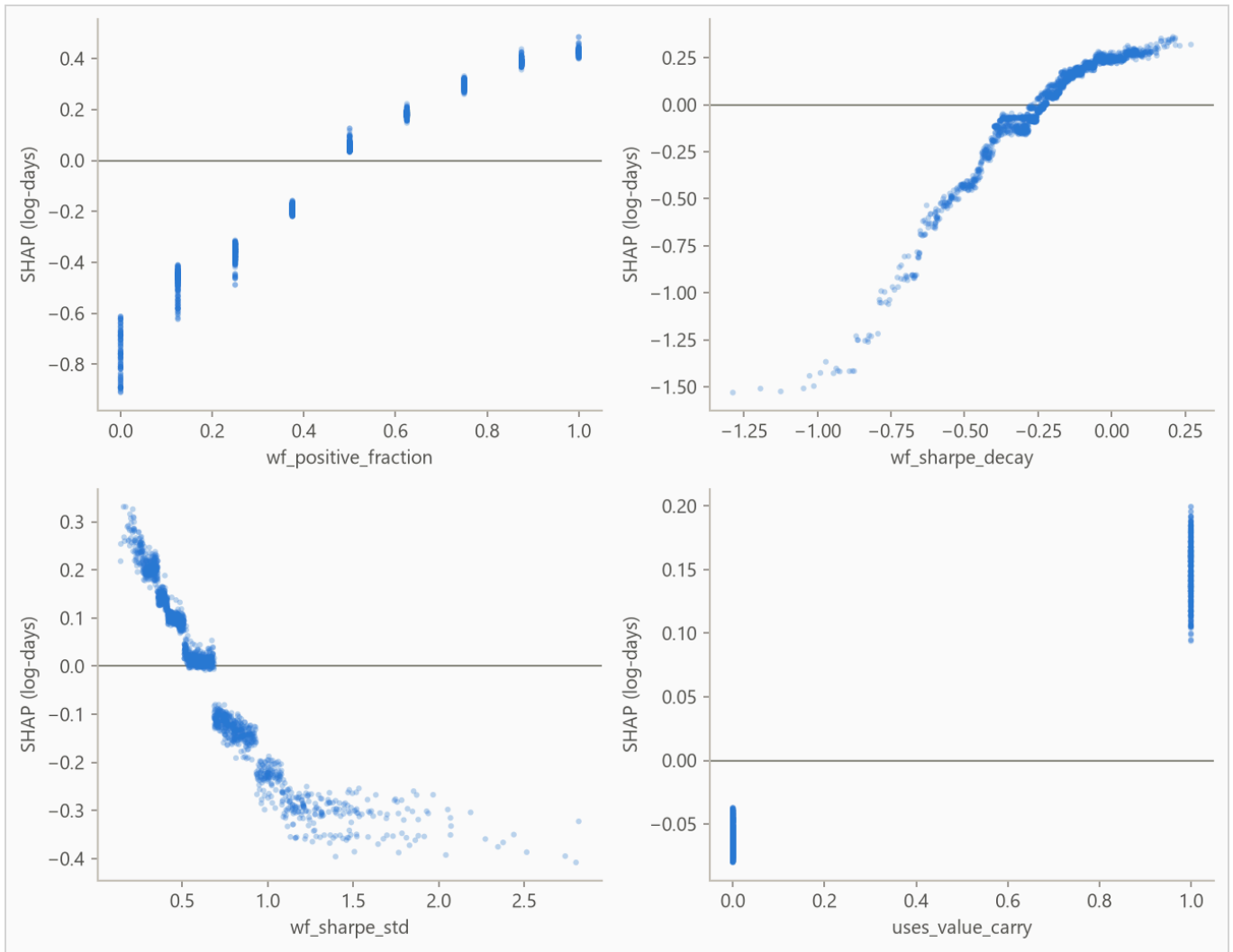
**Figure 3.** Mean absolute attribution by feature. The three walk-forward consistency statistics carry more attribution than every other feature combined.

**Table 5.** Top eight features by mean absolute attribution.

| FEATURE                 | MEAN  ATTRIBUTION |
|-------------------------|-------------------|
| wf_positive_fraction    | 0.294             |
| wf_sharpe_decay         | 0.214             |
| wf_sharpe_std           | 0.138             |
| uses_value_carry        | 0.087             |
| regime_concentration    | 0.081             |
| asset_crypto            | 0.075             |
| log_n_candidates_tested | 0.075             |
| uses_volatility_premium | 0.056             |



**Figure 4.** Per-strategy attributions. High walk-forward positive fraction pushes predicted survival up; a steeply negative walk-forward Sharpe slope pushes it down hard, with the left tail reaching about -1.5 log-days, a factor of roughly 4.5 shorter; high fold-to-fold Sharpe dispersion is penalized. All three directions match the generative design, where consistency rises with true edge and falls with overfitting.



**Figure 5.** Attribution against feature value. The positive-fraction effect is monotone and close to linear, with vertical bands at the nine possible fold fractions and a swing of about 1.3 log-days end to end. The Sharpe-decay effect saturates once the slope turns positive, which is sensible behaviour given that positive decay slopes are mostly noise. The dispersion effect is a descending staircase that flattens past 1.5, where additional dispersion no longer distinguishes anything.

Validation Sharpe does not appear in the top twelve features. On unconditional data that would be strange. Here it is the signature of selection. Every strategy cleared the same threshold, so the metric's remaining variation is mostly overfitting plus noise, and the model routes its attention to walk-forward consistency instead. This is the same phenomenon as the 0.410 in Table 1, seen from inside the model.

Because the generating process is known, this ranking can be checked against the mechanisms actually built in; Addendum A does so.

Directions in that table are more trustworthy than magnitudes. The three walk-forward statistics are correlated proxies for the same latent quantity, and how attribution divides credit among correlated features

reflects the model's internal choices as much as the data. The supportable reading is that the model uses positive fraction about twice as much as Sharpe dispersion, not that positive fraction is twice as important. The feature-family and asset-class effects enter the generator additively and independently, so any reasonable model recovers them; recovering the proxy structure is the informative result.

## 8. Limitations

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**The generator encodes a prior, not evidence.** A model that recovers structure placed there says nothing about whether real strategy metadata contains comparable signal. This work validates a methodology, and the concordance reported here should be treated as an upper bound on production performance.

**Lifetimes are treated as independent.** Real strategies stop working together when a regime breaks. The generator omits that dependence and the bootstrap resamples strategies independently, so the intervals in Table 1 understate real uncertainty and are best read as a lower bound on it.

**Administrative retirement is modeled as independent censoring.** In a live book, strategies are retired because someone observed early decay, which makes the censoring informative and biases survival estimates upward. Handling it properly requires a competing-risks treatment, which is recorded as future work rather than implemented.

**Generator-seed dependence** is examined in Addendum A.

**Every row shares one curve shape.** The predictive scale is a single fitted number, so the model shifts each row's survival curve earlier or later but never widens or narrows it, and two rows with the same predicted median receive identical probabilities at every horizon. Letting the width vary by row would need a model that predicts it, which is future work.

**Harrell's concordance is biased under heavy censoring.** The final fold is 39% censored, where an inverse-probability weighted concordance would be preferable. Fold-to-fold comparisons in Table 2 should be read with that in mind.

## 9. Intended use

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The model is a capital-allocation and review-scheduling prior, not a trade signal. It ranks newly discovered strategies by expected working life and produces a survival curve for each. Predicted median lifetime sets initial position sizing and the first review date; the curve gives a decay schedule against which actual performance can be compared.

Before it informs live allocation, the same temporal cross-validation with the same label re-censoring has to be repeated on production metadata, and the resulting concordance is expected to be lower with wider intervals. No production metadata is presented in this report. The pipeline is built and verified against known ground truth so that a production run needs no methodological work.

The real-data path itself already exists rather than being planned. The same pipeline fits and evaluates any right-censored duration CSV ( `scripts/run_fit_evaluate.py` ), saves both fitted models, and scores new rows later ( `scripts/run_predict.py` ). The repository includes a demonstration on 262,763 real City of Chicago business licences ( `reports/chicago_demo/` ), where the checks that depend on ground truth are absent and the generated report says so.

Nearer-term work, in the order it would be done: a multi-seed sweep to separate data variance from model variance, an inverse-probability weighted concordance alongside Harrell's, a block bootstrap by discovery month to stop understating interval width, and a competing-risks treatment of administrative retirement.

## 10. Reproducing these results

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Python 3.11 or later. From the repository root:

```
pip install -r requirements.txt
python -m pytest
python scripts/run_synthetic_pipeline.py
```

The test suite is 100 tests covering generator invariants, fold and label leakage, and metric behaviour. The pipeline script regenerates the data at seed 7, runs the 5-fold temporal cross-validation, writes `reports/metrics.json` and all figures in about two minutes, then rebuilds this report from that metrics file, so every number here is traceable to a single run rather than transcribed. `requirements.txt` pins the exact dependency versions these numbers were produced with.

The seed-dependence reading in Addendum A is generated from `reports/metrics_seed8.json`, produced by `python scripts/run_synthetic_pipeline.py --seed 8` and saved under that name; the builder reads it when present and states the single-seed limitation when it is not. Building the report also prints `strategy_survival_report.pdf` headlessly when Chrome is available.

## Addendum A. Synthetic ground truth

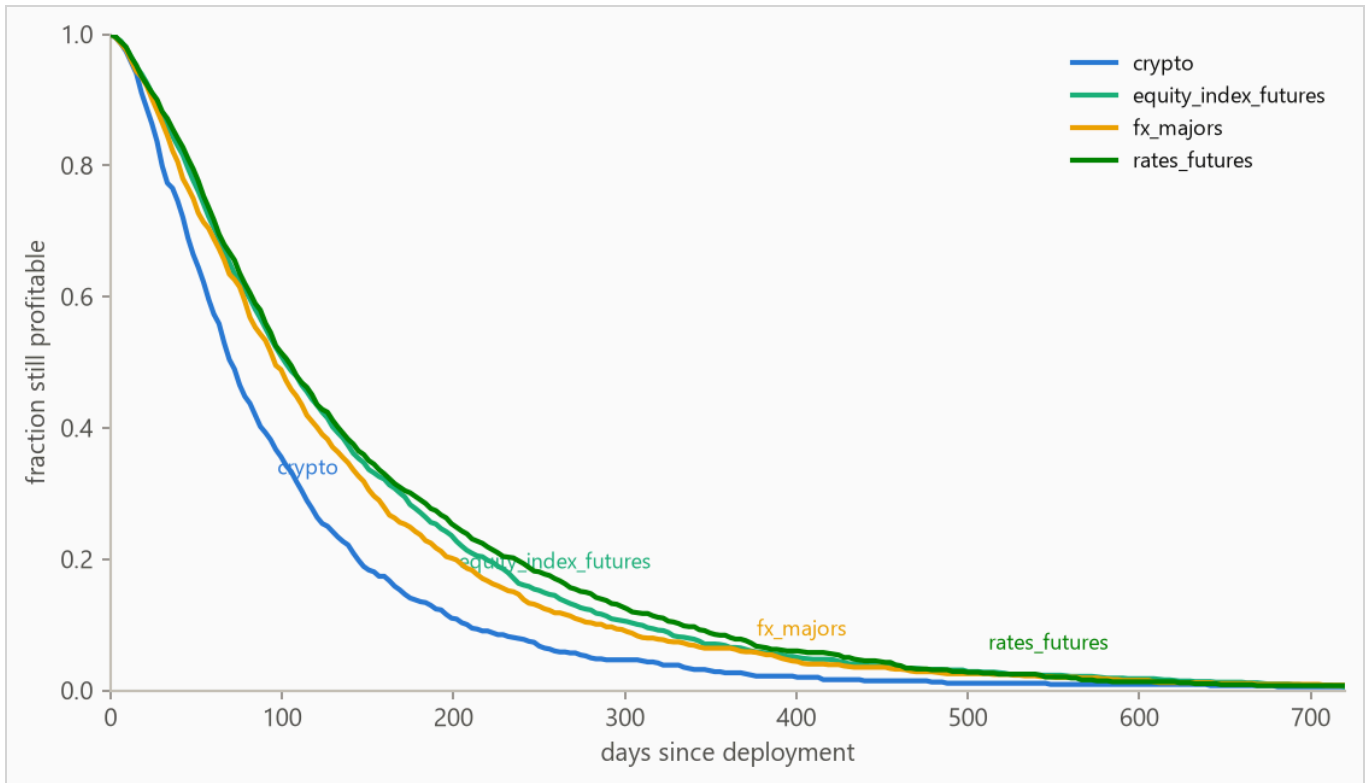
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In this report, all results come from synthetic data. The generator reproduces the statistical structure of a strategy-search pipeline, including selection bias, walk-forward statistics, regime exposures and censored lifetimes, without containing anything proprietary.

Confidentiality is one reason. Verification is the more important one. Because the generating process is known, two checks become available that real data cannot support. The best achievable score can be computed exactly, which bounds any claim about model quality. And the model's feature attributions can be compared against the mechanisms actually built in, which turns interpretation into a test with a right answer. The test suite enforces the first of these. One test asserts the model never outscores the oracle, since beating perfect information would indicate a leak.

### A.1 What the generator builds

The generator writes structure into the population on purpose, so that recovering it is a test rather than an interpretation. Each strategy uses one to three of six feature families, and each family shifts log survival time by a fixed amount, with microstructure the most fragile and value-carry the most durable. Each strategy belongs to one of four asset classes. Crypto is built to decay fastest, a shift of  $-0.30$  on log time against the fx-majors baseline, and it is drawn for roughly 15 percent of rows, so the model has to recover the class effect from a minority slice. Survival time is log-normal around these effects with a scale of  $0.55$  on log-days. Figure 6 shows the resulting survival curves: the crypto curve separates early and stays separated, which is the coarsest piece of built-in structure a fitted model should find.



**Figure 6.** Kaplan-Meier survival curves by asset class. Crypto strategies decay faster by construction, and the model has to recover that from roughly 15% of rows.

## A.2 The oracle ceiling

The oracle ranks strategies by the latent log-time quantity the generator actually used, so its concordance of 0.820 bounds every model; the distance between it and a perfect score is measurement noise, not missing skill. A model reading only the observable metadata faces a ceiling somewhat below the oracle's, since the observables are noisy proxies for the latent quantity.

## A.3 Attribution against the mechanism

The attribution ranking is a test rather than a description, because the generating process is known. The three walk-forward statistics that dominate are not inputs to survival time in the generator. They exist only as the least noisy observable proxies for the latent split between real edge and overfitting, which is the actual driver. The model was not handed that relationship; it found the proxies and leaned on them. Feature-family and asset-class effects return with the signs and rough ordering written into the generator, and the search-intensity penalty appears where a multiple-testing argument predicts.

Two qualifiers keep that reading honest. The walk-forward statistics were assigned generous signal-to-noise when the generator was written, so their dominance was likely before any model ran. And the attributions are



computed on the final model's own training rows, so the check is a qualitative match against the mechanism, not out-of-fold evidence.

## A.4 Generator-seed dependence

The pipeline was rerun end to end with a second seed (seed 8, saved as `reports/metrics_seed8.json`). It selected the same hyperparameters and scored 0.781, inside the seed-7 interval of 0.772 to 0.790. Ranking by validation Sharpe stayed inverted at 0.398, and the oracle ceiling came out at 0.812. The same three walk-forward statistics dominate attribution. Their order matches as well, but the closest pair is separated by a gap smaller than the attribution drift this pipeline has measured across platforms, so the matching order reads as a tie rather than a confirmation. Two seeds is a consistency check rather than a variance estimate. A proper multi-seed sweep remains future work, and the relationships between the numbers in this report deserve more confidence than their individual decimals.

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Generated from `reports/metrics.json` by `scripts/run_build_report.py`. Seed 7, 5,000 strategies, 3,000 out-of-fold test strategies.